

Generation Interconnection System Impact Study Report

For Queue Project Q20-1426

Bolivar 100 kV

June 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1426. The Interconnection Customer proposes a 126.5 MW solar photovoltaic facility interconnecting at the Bolivar 100 kV substation in the Ohio area. The request was received on March 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 483 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,757,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Carbon Ctr 500/345 kV Transformer Bank	Blackrock – Shawville Sta 345kV	113.9	1,998	6.2
Carbon Center – Shawville Sta 500kV Line	Blackrock Substation – Shawville Sta 345kV	107.3	2,111	6.2
Orchard Rd – Shawville Sta 500kV	Scott – Sawville Sta 345kV L	104.7	2,944	7.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carbon Center 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$14,364,000	2% (\$241,000)
Carbon Ctr – Shawville Station 500kV Line	Reconductor line	\$43,413,000	2% (\$729,000)
Orchard Rd – Shawville Sta 500kV line	Reconductor line	\$52,099,000	2% (\$787,000)
	Total allocated to project		\$1,757,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.